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Department of Artificial Intelligence and Machine Learning

SignSync: A Pose-Based Graph-Transformer Framework for Continuous Sign Language Translation

Deep Learning CA3 Report

Submitted By:

Kashish Chelwani

PRN: 24070126082

1. Brief Title

SignSync: A Pose-Based Graph-Transformer Framework for Continuous Sign Language Translation

2. Abstract and Keywords

Continuous Sign Language Translation (SLT) involves aligning the sequence of signs from the video recording to English sentences. This is very challenging because American Sign Language (ASL) conveys information via hand movements, body positioning, facial expressions, and spatial configurations. In this paper, we introduce SignSync, which is a pose-based deep learning model for translating continuous ASL to English using the How2Sign dataset. We encode the clip using 133 keypoints per frame captured via RTMPose; out of the 133 keypoints, 69 keypoints are divided into four skeleton graphs (body, left hand, right hand, and face), which are encoded using spatial temporal graph convolutional network (ST-GCN). Four feature streams are concatenated and projected to 768 dimensions, followed by the mT5-base encoder-decoder that generates English sentence using beam search.

The system is designed as one pipeline using PyTorch including data preparation, model building, training, tuning of hyperparameters and decoding, and evaluation. The metrics used for evaluation are BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, and Word Error Rate (WER). The model was trained using 31,165 clips of How2Sign dataset for 20 epochs and was evaluated on 2349 test clips with results of BLEU-4 = 13.75 and ROUGE-L = 30.66. Control experiments where one body part was removed or frame order shuffled evaluate the importance of each component: removal of the right hand results in BLEU-4 = 2.46, while shuffling of frames gives BLEU-4 = 0.66.

Keywords: Continuous Sign Language Translation, American Sign Language, How2Sign, RTMPose, Spatial-Temporal Graph Convolution Network, mT5, Pose-based Translation.

3. Introduction

Sign language is a language in itself that consists of the coordination of hand signs with body and facial expressions used to communicate messages. In cases where there is no common

language between the signer and non-signer, communication becomes difficult. Sign language translation involves the transformation of continuous signing video into text messages.

The translation of continuous sign languages is a complex problem compared to classification of individual signs due to the complications involved in overlapping of signs, ambiguity in gestures' boundaries and also possible difference in the grammar and syntax of sign languages and English. This implies that the learning process requires visual and linguistic information.

How2Sign is a multimodal dataset for continuous ASL-to-English translation with abundant visual data. Nevertheless, there are still substantial challenges with the task of translation, which has been demonstrated by the results reported as state-of-the-art.

This problem is solved using the SignSync system in a pose-based way. Body, hands, and face keypoints are converted into graphs and then encoded using ST-GCNs. An mT5 Transformer model is used for generating the English sentences. The pose-only system is more lightweight compared to the one that processes all video frames. It does not require storing any video data of the signer and is less sensitive to lighting conditions..

3.1 Problem Statement

Sentence-level English translation of ASL videos into written English using the key points of the body, hands, and face in graph form, and assessing the accuracy and dependence on key points used in the translation.

3.2 Objectives

- Concentrate on the signs found throughout the entire sentence level rather than those found at the isolated lexical level.
- Encode each clip with skeleton graphs (Body, Left Hand, Right Hand, Face) in ST-GCN.
- Relate the extracted graph features to a pre-trained language model (mT5-base) for generating the English sentence.
- Tune the hyperparameters and decoding configurations on the validation set of How2Sign dataset.
- Evaluate the system on the test set of How2Sign using BLEU-1 to BLEU-4, ROUGE-L, and WER scores with confidence intervals.

- Perform control experiments to see if the model uses the hands, the body, the face, and the order of frames.
- Present a working demonstration in the form of a notebook, a web application, and a repository.

3.3 Proposed Contribution

The SignSync system is an engineering and evaluation effort based on preexisting elements. SignSync makes contributions to computer vision and linguistics in terms of a small, commented pipeline implemented in one PyTorch file combining RTMPose keypoints, four ST-GCN part encoders, and mT5-base. In addition, there is training and fine-tuning process, test split evaluation using bootstrap confidence intervals, control experiments demonstrating the contribution of each part of the body, and error analysis of repetition cycles and sentence lengths.

4. Literature Review

Duarte et al. have created How2Sign, a multimodal dataset that includes more than 80 hours of ASL videos along with the corresponding English translations and other modalities [1]. This dataset covers a larger instructional range compared to small sign language datasets; nonetheless, the wide variability in terms of signing as well as sentence-level translation makes the learning task more challenging.

Camgoz et al. have developed Sign Language Transformers by using Transformer-based sequence modeling in order to recognize and translate sign languages [2]. This research highlights the importance of attention-based sequence models for translating visual sequences into language. The Transformer model has been introduced by Vaswani et al. [3].

Yan et al. proposed ST-GCNs for skeleton-based action recognition [4]. Unlike previous techniques that use each joint independently, the technique utilizes a graph representation to model the connections in the human body, while temporal convolutions help to understand the motion dynamics. Action recognition does not include any translational operations; however, this representation is relevant for SignSync's landmark stream.

An initial, gloss-free How2Sign baseline using a Transformer model in conjunction with I3D video representations was proposed by Tarrés et al., attaining a BLEU score of 8.03 [5]. It

becomes clear from their findings that it is essential to consider the intrinsic difficulty of the dataset and the importance of proper analysis of the metrics used for automatic translations. Arib et al. proposed the SignFormer-GCN model which combines Transformers with spatio-temporal graph convolution and obtains a BLEU-4 score of 8.53 on How2Sign [6]. It may be considered a close architectural baseline for SignSync although the results should be directly compared based on matching data splits, preprocessing, and metrics.

4.1 Comparative Analysis of Existing Research

Table 1. Comparative analysis of relevant studies concerning SignSync.

Author(s) & Year	Approach	Key Contribution	Limitation / Scope
Duarte et al. (2021) [1]	How2Sign dataset	Introduced large scale multi-modal dataset for continuous American Sign Language (ASL) with English translations along with additional modalities.	Not a translation model but a dataset source.
Camgoz et al. (2020) [2]	Sign Language Transformers	End-to-end recognition and translation of sign language using Transformer models.	Task design and evaluation are different from the gloss-less How2Sign paradigm.
Yan et al. (2018) [4]	Spatial-Temporal Graph Convolutional Network (ST-GCN)	Represents spatial relations among skeletal joints and their development through time.	Developed originally for skeleton-based action recognition and not for sentence translation.
Tarrés et al. (2023) [5]	Transformer with I3D visual features	Provided an unpolished translation starting point for How2Sign.	Translation quality is still a problem; comparisons require evaluation procedure.
Arib et al. (2025) [6]	SignFormer-GCN	Uses graph convolution and transformer modeling for sign language translation.	Other techniques for multimodal fusion still deserve investigation.
Duarte et al. (2021) [1]	How2Sign dataset	Introduced large scale multi-modal dataset for continuous American Sign Language (ASL) with English translations along with additional modalities.	Not a translation model but a dataset source.
Xue et al. (2021) [8]	mT5	Multilingual pre-trained text-to-text Transformer.	Text only; needs a visual or pose front Text only; needs graphic design or a front end..

4.2 Research Gap

While existing systems provide the score obtained by the system, they do not tell us what exactly the model uses. SignSync solves this problem by adding the use of graph encoders along with a pre-trained language model, showing the score on the test split along with the confidence intervals, and conducting control tests to demonstrate the effect of the hands, body, face, and the order of the frame. This study should be regarded as an experiment, not a novel approach.

5. Methodology

5.1 Dataset Description

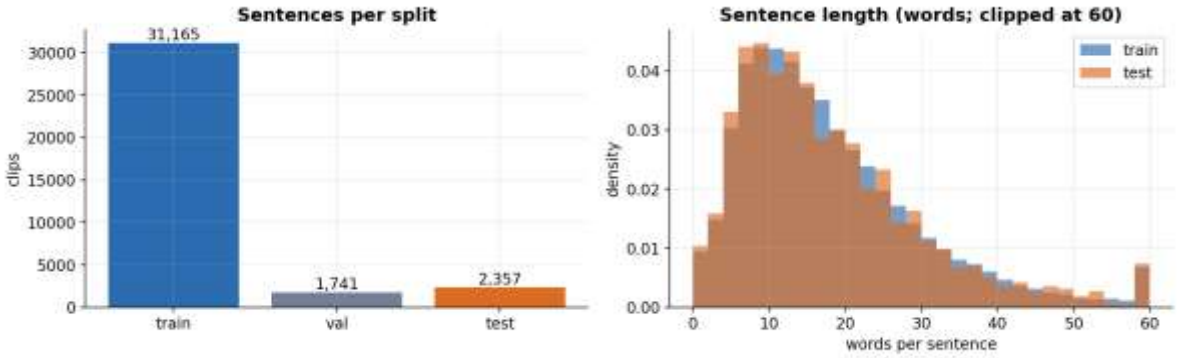


Figure 1. Clips per split and sentence-length distribution of the How2Sign train and test splits.

The dataset picked for this experiment is How2Sign which is a huge scale multimodal dataset for continuous ASL recognition and translation. This dataset has more than 80 hours of video data, English translations, gloss annotations and multimodal data [1]. It contains approximately 35,000 sentences and a vocabulary of 16,000 words (figures mentioned from the dataset's paper itself). For this project, only videos that are at the sentence level from the front view point of view along with the English translated file are considered.

fficially, there are 31,165 training clips, 1,741 validation clips and 2,357 test clips. The RTMPose keypoints for all the clips have been already extracted and hence no poseprocess

takes place within the pipeline. The metrics used for evaluation in the end are BLEU-1 to BLEU-4, ROUGE-L and Word Error Rate (WER).

5.2 Preprocessing and Feature Extraction

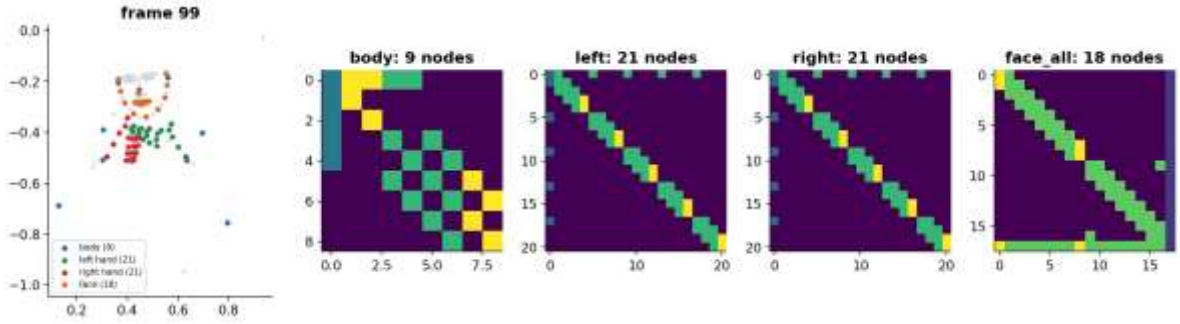


Figure 2. Sample pose frame split into four part graphs, with the adjacency matrix of each graph.

- Keypoints: Each frame contains 133 keypoints from the COCO-WholeBody dataset (17 for the body, 6 for the feet, 68 for the face and 21 per hand), each of which includes an x coordinate, a y coordinate and a confidence score.
- Graphs: 69 keypoints are retained and divided into four graphs: body (9 keypoints), left hand (21), right hand (21) and face (18). Keypoints related to feet and the remaining face points are ignored.
- Frame limit: A maximum of 256 frames per clip is retained, and longer clips are randomly sampled to be shorter (random frames during training and evenly distributed during evaluation).
- Normalisation: Each part is normalised using a reference specific to it (body using shoulders and each hand using wrist to reduce the effect of signer's position and distance from the camera).
- Body information: The body feature is concatenated to the hand and face branches to retain the positional information lost by per-part normalisation.
- Text: The English sentences are tokenised using the mT5 tokenizer with the maximum target length of 64 tokens and "Translate sign language video to English:" prompt preceding the pose features.

5.3 Model Architecture

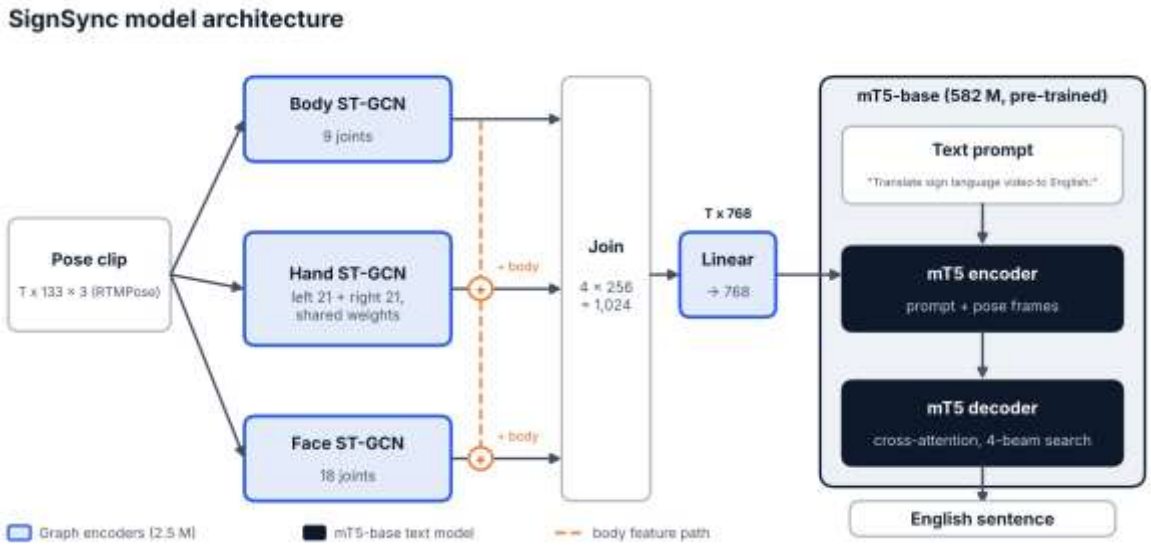


Figure 3. Proposed SignSync pose-based Graph-Transformer architecture.

In the case of the pose branch, the anatomical edges between the joints of each limb are created, and then three ST-GCN blocks with 64, 128, and 256 channels are applied, wherein each block comprises a spatial graph convolution using the normalised adjacency matrix, temporal convolution with kernel size 5, batch normalisation, dropout, and residual connections. Both hands use one set of weights. The node features for each joint are averaged, resulting in four 256-D vectors per frame, which are concatenated to 1,024-D and reduced to 768-D through a linear layer.

The predicted frame vectors are given as input to the mT5-base encoder, based on the text prompt embedded in the text, while the autoregressive decoder of mT5 generates English tokens using cross-attention on the encoded sequence. Graph encoders have 2,465,362 parameters (approximately 2.5 million), while mT5-base has 582 million parameters, so the total parameter size is 585 million, out of which graph encoders constitute less than one percent.

Conceptual process flow: Continuous ASL video → RTMPose 133 keypoints → four part graphs → ST-GCN encoders → joined and projected features → mT5 encoder → mT5 decoder → English sentence.

5.4 Training Procedure

The model was implemented using PyTorch and the mT5 model from Hugging Face library. Training is done using the AdamW optimizer (betas 0.9 and 0.999), peak learning rate 3e-4 with 5% warm-up and then cosine decay schedule, weight decay 1e-4, dropout 0.1, batch size 8 and gradient clipping 1.0, training is conducted in bfloat16 format on GPU. Loss function is token-level cross-entropy loss between generated and reference sentence computed by mT5 ignoring padded tokens. This is the default set of parameters provided in the code.

Training was conducted for 20 epochs using NVIDIA RTX A4000 GPU in 15 hours and the checkpoint with the best BLEU-4 score on validation was saved. Hyperparameters can be tuned using a small random search (6 trials, 3 epochs each on about 500 clips) on learning rate, weight decay, dropout and batch size and the decoding hyperparameters using a grid search on beam width, repetition penalty and no repeat n-grams.

Implementation provides setup, preparation, smoke-test and evaluation scripts, executed notebook, FastAPI web application and Colab demo.

5.5 Evaluation Metrics and Protocol

The model is evaluated using the following evaluation metrics:

BLEU-1 to BLEU-4: Evaluates how similar the generated text is compared to the reference text by calculating the n-gram overlap in corpus BLEU with sacreBLEU (13a tokenisation, case sensitive). High value implies large overlap.

ROUGE-L: Calculates the longest common subsequence of generated text and reference text. Higher value implies more overlap of subsequences.

Word Error Rate (WER): Estimates the number of substitutions, deletions, and insertions that have occurred compared to the reference text in lowercase and without punctuation. Lower value implies small error rate; values higher than 100% are possible for outputs that are larger than the reference text.

Confidence interval is estimated using bootstrap sampling of the test sentences. The control experiment sets the pose of one of the parts (left hand, right hand, both hands, body or face) to

zero or shuffles the frame ordering on the same set of test videos. The fraction of the generated outputs with three-word repetition is the repetition-loop rate.

6. Results and Discussion

The results of How2Sign can be used as the starting point for evaluating SignSync. As per Tarrés et al., the BLEU score is 8.03 [5], and BLEU-4 score according to Arib et al. is 8.53 [6]. In order to make a fair comparison between SignSync and these works, we need to ensure the same data partition, preprocessing method, and metric and decoding scheme. The values within the square brackets are the actual values obtained after the run.

6.1 Quantitative Results

Table 2. Quantitative results on the How2Sign test split.

Metric	Performance Achieved	Interpretation
BLEU-1	38.51	Word overlap; helpful for general content word coverage.
BLEU-2	25.50	Overlap of two-word phrases.
BLEU-3	18.35	Coincidence of three-word sequences.
BLEU-4	13.75	Headline score; relative to published How2Sign baselines.
ROUGE-L	30.66	Measures sequence similarity.
WER	89.4%	Word-level errors; a low score is best.

Table 3. Point and interval estimates of SignSync performance (95% bootstrap interval).

Metric	Achieved value	Possible range
BLEU-1	38.5	37.6–39.4
BLEU-2	25.5	24.6–26.4
BLEU-3	18.4	17.6–19.2
BLEU-4	13.7	13.1–14.5
ROUGE-L	30.7	29.8–31.5
WER (%)	89.4	87.6–91.0

The translation of the entire test split was completed in 451 seconds, or 0.19 seconds per clip. The BLEU-4 on a smoke test using 20 clips was 15.07, while that for 500 clips of the test set used to conduct the control tests was 13.89, which is comparable to the entire split. There were

very few exact matches (32 out of 2,349 clips or 1.4%). 674 translations (28.7%) included a three-word repeat phrase.

6.2 Comparative Study

Table 4. Comparison of SignSync with published How2Sign results.

System	Architecture / setting	Reported or measured BLEU-4	Status
Tarrés et al. [5]	I3D video features + Transformer on How2Sign	8.03	Published; verify metric protocol before direct comparison.
SignFormer-GCN [6]	ST-GCN keypoints with Transformer	8.53	Published.
YouTube-ASL	Large-scale pre-training, then fine-tuning	12.39	Published; value from project notes.
SignSync	RTMPose, four ST-GCNs, mT5-base	13.75	Measured; 2349 test clips.

BLEU-4 is stringent, which means that a sentence with the correct meaning but with different words will receive an almost zero score, which is the reason why even good sign translation systems have much lower scores compared to text-to-text translation. The scoring was reproduced within around 0.75 BLEU-4 from the published score when using poses only, which is less than the bootstrapping interval width, and the reason for the remaining difference is not identified. It is higher than the previously established baselines based on keypoints and I3D scores.

6.3 Ablation Trends

In order to validate the authenticity of the score, the components in the postures were set to zero or the order of the frames was randomized. Table 5 explains the objective and the trend for each control while Table 6 contains the observed scores.

Table 5. Control tests: purpose and observed trend.

Variant	Purpose	Trend
Full model	Reference for the subset	BLEU-4 13.89, consistent with the full test split.
Left hand zeroed	Tests the non-dominant hand	Moderate loss; the right hand carries most of the signal.
Right hand zeroed	Tests the main hand	Largest loss of any single part; BLEU-4 falls to 2.46.
Both hands zeroed	Tests the manual signal	Almost no correct output; BLEU-4 0.77.
Body zeroed	Tests arm and shoulder position	Large loss; body position is needed to place the hands.
Face zeroed	Tests non-manual signals	Clear loss; facial information adds to the translation.
Shuffled frames	Tests temporal order	Collapses to 0.66; the model uses the order of frames.

Table 6. Component-wise control test results (measured on 500 test clips).

Variant	BLEU-1	BLEU-4	ROUGE-L	WER (%)	BLEU-4 lost
C1 Full model	39.26	13.89	30.88	88.0	0.00
C2 Left hand zeroed	36.72	11.72	27.52	90.0	2.17
C3 Right hand zeroed	19.62	2.46	12.68	109.2	11.43
C4 Both hands zeroed	13.91	0.77	7.68	112.5	13.12
C5 Body zeroed	25.43	5.30	17.24	106.0	8.59
C6 Face zeroed	32.75	9.69	23.96	98.6	4.20
C7 Shuffled frame order	15.15	0.66	7.09	126.8	13.23

The output fails if there are no hands or wrong frame ordering, so the translation process relies on the signing rather than solely on the language model. The right hand seems to be far more important than the left one, possibly because the dominant hand performs the main signing action; however, this hypothesis is not proved. Body and face contribute significantly to the output, which justifies the usage of four graphs rather than two hands. WER greater than 100% in a control condition indicates that the outputs were longer than references due to repetitions.

6.4 Qualitative Outcomes

Qualitative assessment consists of evaluating English sentences generated by the model against the corresponding references. The reason is to understand if the model preserves the topic, keywords, order, names, numbers, and overall meaning of the sentence. Samples from the test split:

- Exact: "Watch my fingers as I do this." and "We're going to do two ounces of gin." are copied literally.
- Close: reference "Make sure you get a nice good crease there." results in "Make sure you have a nice good crease there."
- The same idea but different wording: reference "And it doesn't tangle up." becomes "It doesn't get stuck anywhere."
- Failure: reference "That is what your Cafe de Paris is going to look like." turns into "Did you know that?"
- Loop: reference relating to taking dirt from the walk space translates to "I'm going to go over here and I'm going to go over here ..." and so on.

As Table 7 shows, the quality score is correlated with the length of a sentence. Very short sentences receive low scores due to one incorrect word breaking all n-grams, and very long sentences are hard to replicate entirely.

Table 7. BLEU-4 by reference sentence length.

Reference length (words)	Clips	BLEU-4
1 to 5	280	4.84
6 to 10	501	12.48
11 to 20	819	14.20
21 to 40	614	13.69
41 or more	135	7.72

6.5 Limitations

Some disadvantages of this approach are listed below:

- A loop repetition has an effect on about 29% of output samples, scoring negatively; outputs that contain the loop get BLEU-4 score of around 7 compared to 17 for those who don't have it.
- Only 69 out of 133 keypoints are used, meaning that fine finger position and facial expression information is not accessible to the model.
- Error in pose estimation under fast movement or occlusion directly affects the encoder.
- There are eight test videos without any pose file, thus only 2,349 out of 2,357 were used for the evaluation.
- BLEU and ROUGE-L scoring systems do not measure semantic accuracy correctly.
- Test samples for control are limited to a 500-clip subset, thus slight difference between control samples is not significant.
- Signer metadata is not taken into account, thus signer-invariant performance was not estimated.

6.6 Future Improvements

Future directions for improvement include training on the whole official training set for more epochs, performing decoding only search with beam width, repetition penalty, and no-repeat n-gram size (which targets repetitions), and applying early stopping using validation scores. The RGB network could incorporate finger shapes and facial information, additional facial points, and non-manual signs could be used, while signer-independent evaluation could be employed when signer identities are known. For extending the approach to Indian Sign Language or live-streaming, proper data sets and retraining would be required.

7. Conclusion

This paper introduces SignSync, a system for real-time translation of signs from American Sign Language to English using body, hand, and face keypoints. SignSync makes use of RTMPose keypoints along with four ST-GCN part encoders and an mT5-base encoder-decoder, as a single PyTorch pipeline. When tested on the How2Sign test split, SignSync achieves a BLEU-4 score of 13.75, ROUGE-L of 30.66, and WER89.4%. In control experiments, it is observed that the translation is more sensitive to right-hand keypoints,

followed by the body, face, and left hand. It is also sensitive to proper ordering of frames. About 29% of output translations have repetitions, reducing the score.

8. References

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